

- TP 1 : Introduction to Cloud
- Professor: Ouassim Karrakchou
- Level : 4th year

TP 1

Introduction to Cloud

Objectives:

The goal of this TP is to introduce you to the core concepts of the Cloud

We will do the following tasks:

- Get your access to the AWS Academy and Azure for Students accounts.
- Discover the key architectural components of a Cloud (regions, availability zones, services).
- Discover the basic compute and storage services available in Azure and AWS.

Evaluation:

These TPs have two parts that are evaluated:

1. Experimental part: call the Professor to verify what you did when it tells you to do so.
2. Assignment: an assignment needs to be done at the end of the class. Please send it by email to your TP teacher **before midnight the day of the TP**:
 - a. Awatif ROUIJEL awatifrouijel@gmail.com
 - b. Mohammed SOUIDI souididoc@gmail.com
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The TP must be done in groups of 2 to 3 students. The groups must stay the same for all the next TPs.

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I- Introduction:

The first TP of the Cloud Foundations course will introduce you to the Cloud. You will first get your student access to the Microsoft Azure and Amazon AWS clouds, and follow an e-learning on the official AWS academy platform that introduces you to the Cloud, as well as provide with several labs where you will discover how to configure the AWS Cloud to create several resources. Today, we will focus on learning how the AWS Cloud is organized by watching an e-learning and doing a knowledge check. You will then learn how to create your first service in the AWS Cloud: a virtual machine. You will first do this in a guided manner in AWS, then you will have to search on the Microsoft Azure documentation how to replicate this on Azure. You will notice that several concepts are very similar in different Cloud providers.

II- Get Access to the Cloud:

1. Go to the page: <https://azure.microsoft.com/en-us/free/students>
2. Click on **Start Free**
3. Connect using your **UIR email account**
4. Fill the form with your information. On the first page, you will need to fill the **university name**. **Leave this field at the end**, and when all other fields are filled, put **International University of Rabat** in it, then press **Enter** in your keyboard.
5. Fill the rest of your personal information. Make sure you put your own personal phone number there without any country code (example: 0612345678).
6. If everything goes well, your account should be activated, and you should be redirected to the Azure portal: <https://portal.azure.com/#home>
7. **Call the Professor** to show him/her what you did.
8. The **Professor** will also create you an account on the **AWS Academy portal**.

III- Learn how the Cloud is organized:

1. Connect to the **AWS Academy portal**, and access the **Cloud Foundations course**.
2. Go the **Module 3**
3. Watch the e-learning videos of this module, and do the knowledge check (10 questions)

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4. **Call the Professor** to show him/her the result of your knowledge check.

IV- Create a simple VM in AWS:

1. Go now to **Module 6** in the Cloud Foundations course in the AWS Academy
2. Click on Lab 3, then on Load Lab 3 to open the lab window.
3. Follow the instructions in the lab to create your first VM in AWS.
4. There are 6 tasks in the lab. Whenever you finish a task (except for task 5), **call the Professor** to show him/her what you did.

V- Create a simple VM in Azure:

1. Go now to **the Azure Portal**
2. Your goal is to **replicate what you did in AWS in Azure**. Your goal is then to **create a B1S VM (or equivalent, it should be low cost) in Azure**, install in it a simple **web server**, and **access a test web page** from it.
3. Investigate how to do this by **searching online on the Azure documentation**. Then **do it in your Azure portal**.
4. **Call the Professor** to show him/her what you did.
5. **Delete the VM and all the associated resources** to not consume credits from your Azure for Students account

VI- Assignment:

Based on what you learned in the TP explain the different steps needed to create an Ubuntu VM in Azure and in AWS, and the different resources that are created. Use screenshots from the Azure portal and the AWS console. **For the case of Azure, when you have finished, make sure you delete all the resources you used (show the steps).**